

Name: ..... Maths Class: .....

## SYDNEY TECHNICAL HIGH SCHOOL



### YEAR 11 PRELIMINARY HSC COURSE

#### Mathematics

September 2008

TIME ALLOWED: 120 minutes

#### *Instructions:*

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied at the time of marking

(FOR MARKERS USE ONLY)

| 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | TOTAL |
|-----|-----|-----|-----|-----|-----|-----|-----|-------|
| /14 | /13 | /13 | /12 | /13 | /12 | /12 | /11 | /100  |

**QUESTION 1: (14 Marks)**

- |                                                                                                                                                  | <b>Marks</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| (a) Fully factorise $5x^2 - 45$                                                                                                                  | <b>2</b>     |
| (b) Rationalise the denominator and simplify: $\frac{\sqrt{5}}{\sqrt{5}-1}$                                                                      | <b>2</b>     |
| (c) Simplify $7\sqrt{3} - 2\sqrt{27}$                                                                                                            | <b>2</b>     |
| (d) Expand and simplify $(x-2)(x^2 + 2x + 4)$                                                                                                    | <b>1</b>     |
| (e) The diagram below is of a regular hexagon with a triangle ABC attached made by extending two of its sides.                                   |              |
|                                                                                                                                                  |              |
| (i) Find the size of each of the internal angles of the hexagon                                                                                  | <b>1</b>     |
| (ii) Giving all reasons, identify the type of triangle ABC drawn                                                                                 | <b>3</b>     |
| (f) Find the equation of the line through the point (3, -4) and parallel to the line $x + 2y - 2 = 0$ . <u>Give your answer in general form.</u> | <b>3</b>     |

**QUESTION 2: (13 Marks)**

**Marks**

(a) Find  $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 3}{2x^2 - 5}$  1

(b) If  $\cos \theta = \frac{5}{7}$  and  $0^\circ \leq \theta \leq 90^\circ$  find  $\tan \theta$  as a surd in simplified terms 2

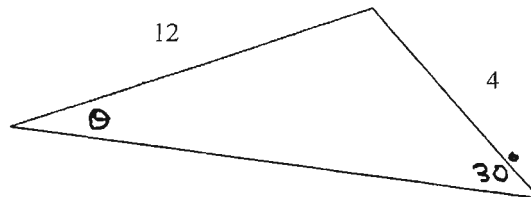
(c) Differentiate the following with respect to  $x$ :

(i)  $y = (4 + x)(2 - x)$  1

(ii)  $y = \frac{3}{x^2}$  1

(iii)  $y = (5x^2 - 4)^4$  1

(d)



(i) Find the value of  $\theta$  to the nearest minute 3

(ii)  $\theta = 170^\circ 24'$  is also a solution to the problem above. Explain why this “solution” is not an answer to the question. 1

(e) Solve  $|2x - 3| \leq 5$  and plot the solution on a number line 3

**QUESTION 3: (13 Marks):**

**Marks**

- (a) Find the point(s) on the line  $y = x^3 - 12x + 1$  where the tangent is parallel to the x-axis **3**

- (b) The functions  $f(x)$  and  $g(x)$  are defined as:

$$f(x) = 1 - x^2 \quad \text{and} \quad g(x) = \sqrt{x}$$

- (i) On a set of neat axes, sketch  $y = f(x)$  showing all intercepts with the co-ordinate axes. **2**
- (ii) Find  $f(-3)$  **1**
- (iii) Find the Range of  $f(x)$  **1**
- (iv) What is the natural domain for  $g(x)$  ? **1**
- (v) Find  $f(g(x))$  **1**

- (c) Differentiate the following with respect to x:  
(give your answers without negative indices)

(i)  $y = \frac{2 - x^2}{x^2 + 2}$  **2**

(ii)  $y = \sqrt{1 - x^2}$  **2**

# **QUESTION 4: (12 Marks)**

Marks

- (a) Given that  $\alpha$  and  $\beta$  are the roots to  $2x^2 - 3x + 2 = 0$ , find the value of the following:

**(DO NOT ATTEMPT TO SOLVE THE EQUATION)**

- |                                                    |   |
|----------------------------------------------------|---|
| (i) $\alpha + \beta$                               | 1 |
| (ii) $\alpha\beta$                                 | 1 |
| (iii) $\alpha^2 + \beta^2$                         | 2 |
| (iv) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ | 2 |

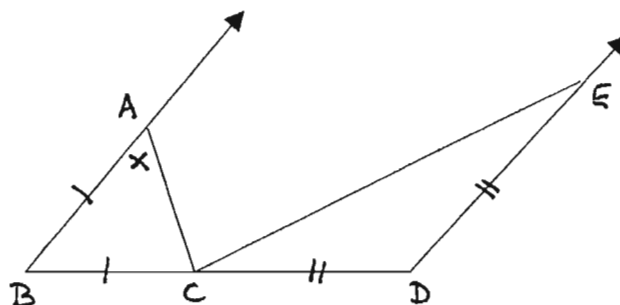
- (b) In the diagram at right,

$AB \parallel ED$

$AB = BC$  and  $CD = DE$

B, C and D are collinear

$\angle BAC = x$



Give reasons for all of your answers to the questions below

- |                                                            |   |
|------------------------------------------------------------|---|
| (i) Find $\angle ACB$ in terms of $x$                      | 1 |
| (ii) Find $\angle ABC$ in terms of $x$                     | 1 |
| (iii) Why is $\angle CDE = 2x$ ?                           | 1 |
| (iv) Hence find an expression for $\angle DCE$             | 2 |
| (v) What can you conclude about the size of $\angle ACE$ ? | 1 |

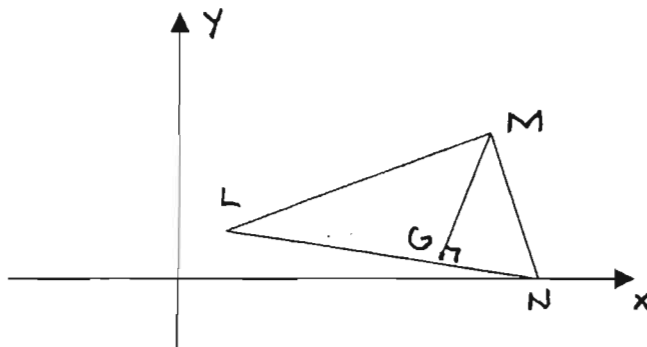
**QUESTION 5: (13 Marks)**

|                                                 |                                                                                                               | <b>Marks</b> |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------|
| (a)                                             | Find the equation of the tangent to the curve $y = x^3 - 2x + 1$ at the point where $x = 2$                   | <b>3</b>     |
| (b)                                             | The point P (2, 2) lies on the curve $y = 2x - x^2 + 2$<br>Find the equation of the normal to this curve at P | <b>3</b>     |
| (c)                                             | (i) Sketch $y =  2x - 1 $ showing all intercepts                                                              | <b>2</b>     |
|                                                 | (ii) Draw another line on your sketch which could be used to solve $ 2x - 1  = 3$ and label it.               | <b>1</b>     |
| <b>THERE IS NO NEED TO SOLVE THE INEQUALITY</b> |                                                                                                               |              |
| (d)                                             | Find the values of $k$ for which the equation $4x^2 - 4(k - 3)x + 1 = 0$ has no real roots.                   | <b>4</b>     |

### QUESTION 6: (12 Marks)

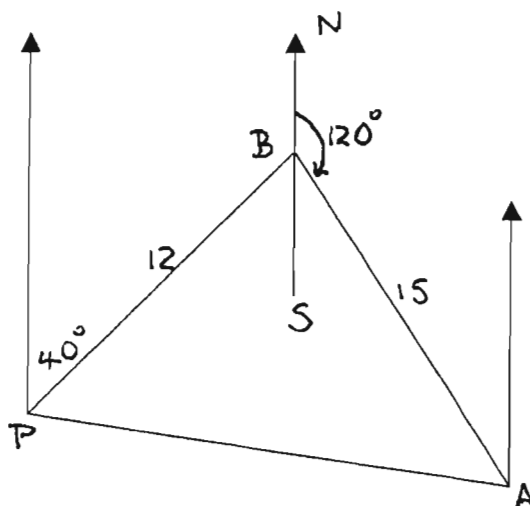
Marks

- (a) The points  $L(1, 1)$ ,  $M(4, 2)$ , and  $N(5, 0)$  shown below form a triangle.  $MG$  is an altitude of this triangle.



- |                                       |   |
|---------------------------------------|---|
| (i) Find the length of $LN$           | 1 |
| (ii) Find the equation of $LN$        | 2 |
| (iii) Find the height $MG$            | 2 |
| (iv) Find the area of $\triangle LMN$ | 2 |

- (b) The diagram below shows the course of a ship, which sails from a port  $P$  on a bearing of  $040^\circ$  for 12 km before changing course to a bearing of  $120^\circ$  and travelling a further 15 km to a destination  $A$ .



- |                                                               |   |
|---------------------------------------------------------------|---|
| (i) Explain why $\angle PBA = 100^\circ$                      | 1 |
| (ii) Find the distance of $A$ from $P$ to the nearest km      | 2 |
| (iii) Find the bearing of $P$ from $A$ to the nearest degree. | 2 |

**QUESTION 7: (12 Marks)**

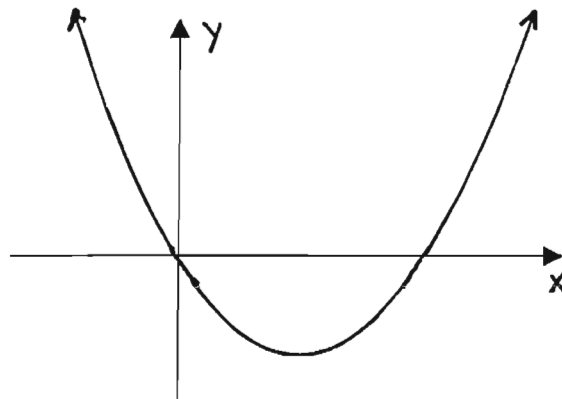
**Marks**

(a) Solve  $5^{1-2x} = 25$  **1**

(b) The point  $(k, 1)$  is equidistant from the points A(2, 3) and B(-1, 4). Find the value of  $k$ . **3**

(c) Solve  $2\cos^2 x - 1 = 0$  for  $0^\circ \leq x \leq 360^\circ$  **4**

(d) The diagram below represents  $y = kx^2 - 4kx$



(i) Find the intercepts with the  $x$ -axis **1**

(ii) Find the co-ordinates of the vertex in terms of  $k$  **2**

(iii) What does the result to part (ii) above imply about the number of roots to the equation  $kx^2 - 4kx + A$  when  $A > 4k$ ? **1**

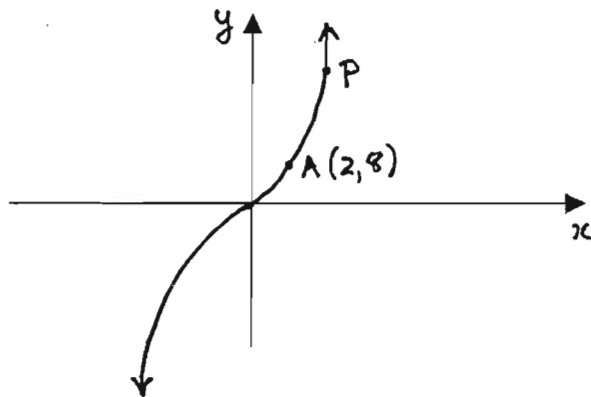
**QUESTION 8: (11 Marks)**

**Marks**

- (a) On a set axes draw the graph of  $(x-1)^2 + (y+2)^2 = 9$  **3**  
and on it shade the region satisfying  $(x-1)^2 + (y+2)^2 \leq 9$

- (b) Show that  $\frac{(1 - \cos \theta)(1 + \cos \theta)}{\sin \theta \cos \theta} = \tan \theta$  **2**

- (c) The curve given below is  $y = x^3$ . The point A on it is (2, 8).



- (i) The point P has an x value of  $2 + h$ .  
Give the y co-ordinate of point P. **1**  
*(You may use the formula  $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ )*
- (ii) Showing all working, prove that the slope of the chord PA is given by **3**  
$$12 + 6h + h^2$$
- (iii) Find the limit of the slope of chord PA as P approaches A along the curve (ie as  $h \rightarrow 0$ ) **1**
- (iv) Give an interpretation of the answer you obtained in part (iii) above **1**

***END OF EXAMINATION PAPER***

Teacher's Name:

Student's Name/Nº:

$$(1) (a) 5(x+3)(x-3) \leftarrow (1)$$

$$(b) \frac{\sqrt{5}(\sqrt{5}+1)}{4} = \frac{5+\sqrt{5}}{4} \leftarrow (1)$$

$$(c) 7\sqrt{3} - 6\sqrt{3} = \sqrt{3} \leftarrow (1)$$

$$(d) x^3 - 8$$

$$(e) (i) \text{ Angle sum} = 720^\circ$$

$\therefore$  Each angle is  $120^\circ$  (1)

$$(ii) \text{ Since } \left. \begin{array}{l} \angle SAC = 180^\circ \\ \angle BAC = 60^\circ \end{array} \right\} (1)$$

Similarly,  $\angle ABC = 60^\circ$

$\therefore \angle ACB = 60^\circ$  (angle sum of  $\triangle ABC$ ) (1)

$\therefore \triangle ABC$  is equilateral (1)

(f) EITHER:

$$\text{Slope} = -\frac{1}{2} \quad (1)$$

$$\therefore y+4 = -\frac{1}{2}(x-3)$$

$$2y+8 = -x+3$$

$$\therefore x+2y+5=0 \quad (2)$$

[subtract 1 mark if not in general form]

OR

$$x+2y+k=0 \quad (1)$$

substitute (3, -4)

$$\therefore 3-8+k=0$$

$$k=5 \quad (1)$$

$$\therefore x+2y+5=0 \quad (1)$$

$$(2) (a) \frac{3}{2}$$

$$(b) \left. \begin{array}{l} \tan \theta = \frac{\sqrt{24}}{5} \\ = \frac{2\sqrt{6}}{5} \end{array} \right\} \text{either } (1)$$

$$(c) (i) -2-2x$$

$$(ii) -\frac{6}{x^3} \text{ OR } -6x^{-3}$$

$$(iii) 40x(5x^2-4)^3$$

$$(d) (i) \frac{\sin \theta}{4} = \frac{\sin 30^\circ}{12} \quad (1)$$

$$\therefore \sin \theta = \frac{1}{2} \quad (1)$$

$$\theta = 9^\circ 36' \quad (1)$$

(subtract 1 if not to nearest minute)

(ii) The angle sum of the triangle would exceed  $180^\circ$

(c) EITHER

$$2x-3 \leq 5 \quad \text{and} \quad 2x-3 \geq -5$$

$$\text{either } \left\{ \begin{array}{l} x \leq 4 \\ -1 \leq x \leq 4 \end{array} \right. \quad \text{and} \quad x \geq -1$$

$$(2) \left\{ \begin{array}{l} -1 \leq x \leq 4 \end{array} \right. \quad (1)$$

$$\text{OR} \quad 2x-3 \leq 5 \quad \text{or} \quad 2x-3 \geq -5$$

$$x \leq 4 \quad \text{or} \quad x \geq -1 \quad (1)$$

$$\left\{ \begin{array}{l} -1 \leq x \leq 4 \end{array} \right. \quad (1)$$

If they use this method they must write  $-1 \leq x \leq 4$  to get the extra mark

Teacher's Name:

Student's Name/N°:

(3)

$$(a) \frac{dy}{dx} = 3x^2 - 12$$

To be parallel with x-axis

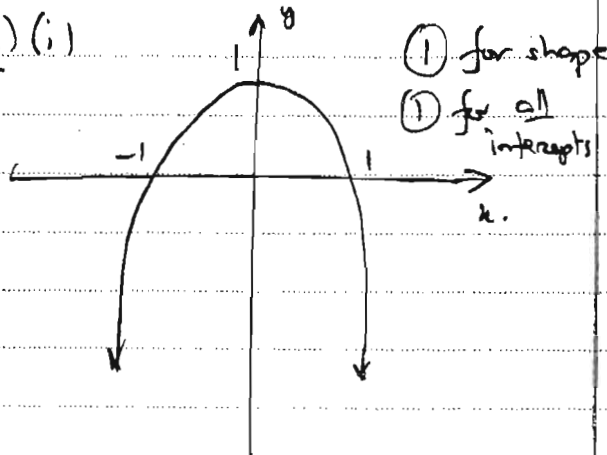
$$\frac{dy}{dx} = 0 \quad \leftarrow (1) \text{ (if you see this angle)}$$

$$\therefore x^2 = 4$$

$$\therefore x = \pm 2$$

Pts are (2, -15) or (-2, 17)

(b) (i)



$$(ii) f(-3) = -8$$

$$(iii) \text{ Range } y \leq 1$$

$$(iv) x \geq 0 \quad (\text{no mark if the equals sign is missing})$$

$$(v) 1 - (\sqrt{x})^2 = 1 - x$$

$$(c) \frac{dy}{dx} = \frac{(x^2+2)(-2x) - (2-x^2)(2x)}{(x^2+2)^2} \quad \leftarrow (1)$$

$$= \frac{-8x}{(x^2+2)^2} \quad \leftarrow (1)$$

$$(ii) \frac{dy}{dx} = \frac{1}{2} (1-x^2)^{-1/2} (-2x) \quad (1)$$

$$= \frac{-x}{\sqrt{1-x^2}} \quad (1)$$

(4)

$$(a) (i) \alpha + \beta = \frac{3}{2}$$

$$(ii) \alpha\beta = 1$$

$$(iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad (1)$$

$$= \frac{9}{4} - 2 \quad (1)$$

$$(iv) \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} \quad (1)$$

$$= \frac{1/4}{1}$$

$$= \frac{1}{4} \quad (1)$$

$$(b) (i) \angle ACB = x^\circ \quad (\text{base angles of isosceles } \triangle ABC)$$

$$(ii) \angle ABC = 180 - 2x \quad (\text{angle sum of } \triangle ABC)$$

$$(iii) \text{ Because it is co-interior to } \angle ABC \text{ and they must sum to } 180^\circ$$

$$(iv) \angle DCE + \angle DEC = 180 - 2x$$

$$\text{and they are equal (angle sum of isosceles } \triangle DCE)$$

$$\therefore \angle DCE = 90 - x$$

$$(v) \text{ It is } 90^\circ$$

(5) (a)  $y = x^3 - 2x + 1$

$$\frac{dy}{dx} = 3x^2 - 2$$

At  $x=2$

$$m_T = 10 \quad (1)$$

$$y = 5 \quad (1)$$

$$y - 5 = 10(x - 2)$$

$$y - 5 = 10x - 20$$

$$y = 10x - 15 \quad (1)$$

(b)  $\frac{dy}{dx} = 2 - 2x$

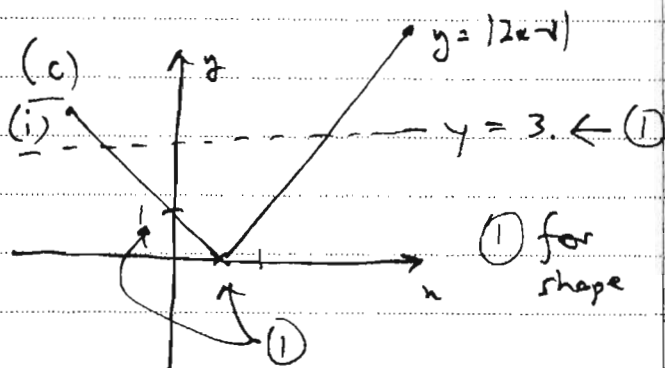
At  $(3, 2)$   $m_T = -2 \leftarrow (1)$

$$m_N = \frac{1}{2} \leftarrow (1)$$

$$y - 2 = \frac{1}{2}(x - 2)$$

$$2y - 4 = x - 2$$

$$x - 2y + 2 = 0 \leftarrow (1)$$



(d) No real roots  $\Rightarrow \Delta = 0 \quad (1)$

$$\therefore [4(k-3)]^2 - 4 \cdot 4 = 0 \quad (1)$$

$$\therefore k^2 - 6k + 9 - 1 = 0$$

$$k^2 - 6k + 8 = 0$$

$$(k-4)(k-2) = 0$$

$$k = 4 \text{ or } k = 2 \quad (2)$$

(6)

(a)(i)  $LN = \sqrt{4^2 + 1^2}$

$$= \sqrt{17} \quad (1)$$

(ii)  $m_N = -\frac{1}{4} \quad (1)$

$$y - 12 = -\frac{1}{4}(x - 1)$$

$$4y - 4 = -x + 1$$

$$x + 4y - 5 = 0 \quad (1)$$

(iii)  $p = \left| \frac{4 + 8 - 5}{\sqrt{17}} \right| \quad (1)$

$$= \frac{7}{\sqrt{17}} \quad (1)$$

(iv)  $\text{Area} = \frac{1}{2} \times \frac{7}{\sqrt{17}} \times \frac{\sqrt{17}}{1}$

$$= \frac{7}{2} \text{ m}^2$$

(b) By alternate angles  $\angle PBS = 40^\circ$

Also, ~~by~~ by straight angles,

$$\angle ABS = 60^\circ$$

$$\therefore \angle PBA = 100^\circ$$

(ii)  $PA^2 = 12^2 + 15^2 - 2 \times 12 \times 15 \cos 100^\circ$

$$PA = 20.77$$

$$\approx 21 \text{ km} \leftarrow (1) \text{ (no penalty if not rounded)}$$

(iii)  $\frac{\sin \theta}{12} = \frac{\sin 100^\circ}{20.77}$

$$\therefore \theta = 34^\circ 41'$$

$$\approx 35^\circ$$

$$\therefore \text{Bearing is } S 85^\circ W \text{ or } 265^\circ$$

Teacher's Name:

Student's Name/N<sup>o</sup>:

7 (a)  $5^{1-2x} = 5^2$

$$1-2x = 2$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

(b)

$$\left. \begin{aligned} d_{PA} &= \sqrt{(2-k)^2 + 4} \\ d_{PB} &= \sqrt{(k+1)^2 + 9} \end{aligned} \right\} \textcircled{1}$$

$$\therefore 4 - 4k + k^2 + 4 = k^2 + 2k + 9$$

$$6k = -2$$

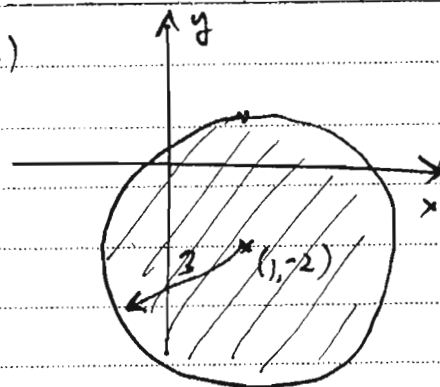
$$k = -\frac{1}{3} \textcircled{1}$$

(c)  $\cos^2 x = \frac{1}{2}$

$$\cos x = \pm \frac{1}{\sqrt{2}} \textcircled{2}$$

$$\therefore x = 45^\circ, 135^\circ, 225^\circ, 315^\circ \textcircled{2}$$

8 (a)



1 for slope

1 for centre and radius

1 for shading

(b) LHS =  $\frac{1 - \cos^2 \theta}{\sin \theta \cos \theta}$

$$= \frac{\sin^2 \theta}{\sin \theta \cos \theta} \textcircled{1}$$

$$= \tan \theta \textcircled{1}$$

(d) (i)  $x=0, x=4$

(ii) V is  $(2, -4k)$

$\textcircled{1} \quad \textcircled{1}$

(iii) there are none

(c) (i) P has y co-ordinate of  $(2+h)^3$  [no need to expand here]

(ii) Slope PA =  $\frac{(2+h)^3 - 8}{2+h-2} \textcircled{1}$

$$= \frac{8 + 12h + 6h^2 + h^3 - 8}{h} \textcircled{1}$$

the mark  $\rightarrow$   
for the expansion  
may come out of part (i)  $= 12 + 6h + h^2 \textcircled{1}$

(iii)  $\lim_{h \rightarrow 0} \text{slope PA} = 12 \textcircled{1}$

(iv) This is the slope of the tangent to  $y=x^3$  at  $(2, 8)$

[OR the derivative to the curve at  $x=2$   
or similar]